

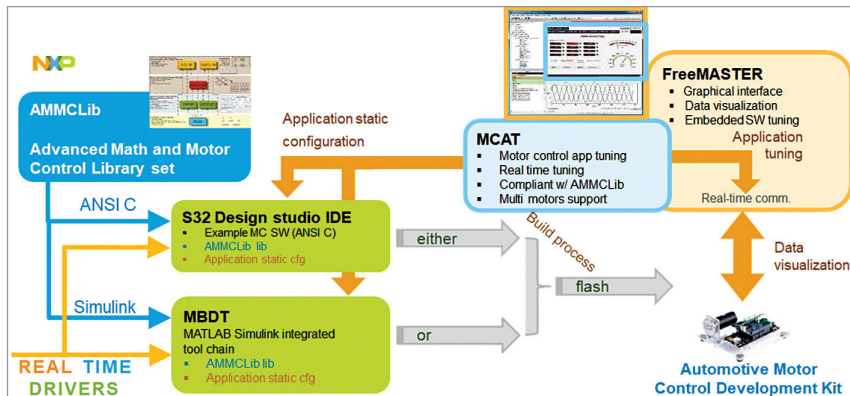


Fig. 6: An example of a seamless overall development flow for electric 2W/3W motor control

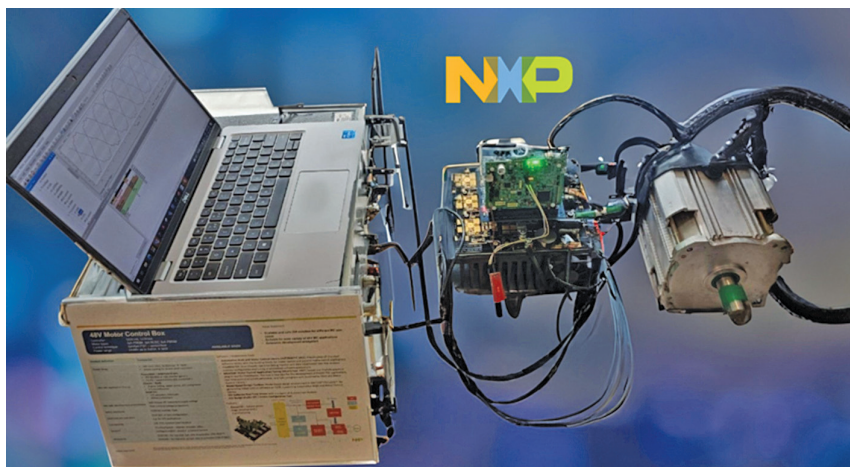


Fig. 7: A benchtop setup for 48V motor control that supports the model-based design

- Graphical control panel with the ability to customise the user interface
- Ability to read and display variables in real time
- High-speed recorded data to simulate a digital oscilloscope
- Real-time tuning for control processes such as current loop, speed loop, sensor/sensorless, etc
- Support for expert modes such as the pole placement method for control parameter estimation

Modern firmware architecture

Developers of motor controllers will benefit from adopting a modern firmware architecture that relies on automotive-quality, production-ready device drivers for the microcontroller. These drivers should be compatible with AUTOSAR and non-AUTOSAR,

keeping both current and future needs in mind.

Overall development flow

Stitching it all together, Fig. 6 shows an overall development flow that is focused on maximising developer efficiency—providing the fastest path to market on the most optimised motor controller solution.

Development of motor controller firmware can be a daunting task at first. However, adopting a modern architecture and development framework that incorporates model-based design, automotive math, and motor control libraries, along with real-time debug tools, can ease the journey and significantly accelerate time to market for startups and established teams alike. **EFY**

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